

Rupam Mukherjee

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Research Interests

Medical Image Computing; Inverse Problems; Machine Learning; Digital & Computational Pathology.

Education

- **Indian Institute of Technology Kharagpur** Jul 2024 – May 2026
M.Tech in Medical Imaging & Informatics; GPA: 8.82/10 Kharagpur, India
 - Coursework: Digital Image Processing, Biostatistics, Physics & Instrumentation of Medical Imaging, Medical Image Analysis, Pathological Image Processing, Molecular Imaging, Computer Vision.
- **Kalyani Government Engineering College** Oct 2020 – May 2024
B.Tech in Computer Science & Engineering; GPA: 8.88/10 West Bengal, India
 - Coursework: Programming, Engineering Mathematics, Data Structures, Algorithms, Machine Learning, Theory of Computation, Compiler Design, Operating Systems.

Publications & Preprints

- **Mukherjee, R., Daniel, R., Hazra, S., Dasgupta, S., Mandal, S.** *RAA-MIL: A Novel Framework for Classification of Oral Cytology*. **Accepted at IEEE ISBI 2026**. [[arXiv Version](#)]
- **Hazra, S., Mukherjee, R., Daniel, R., Dasgupta, S., Mandal, S.** *Uncertainty-Aware Multimodal Fusion for Oral Lesion Classification*. **Accepted at MICCAI MultiTab Workshop 2026**. [[arXiv](#)]
- **Mukherjee, R., Daniel, R., Hazra, S., Dasgupta, S., Mandal, S.** *Multimodal RGB-HSI Feature Fusion with Patient-Aware Incremental Heuristic Meta-Learning for Oral Lesion Classification*. [[arXiv](#)]

Experience

- **Project Associate-1** Jun 2026 – Jul 2026
AI/ML Development CRTDH, IIT Kharagpur
 - Developed a clinically aware Oral Cancer AI Triage System.
 - Coordinated with Software Dev teams for platform development and feature integration.
- **Student Lead — IEEE EMBS BHI 2025 Data Competition** 2025
Team Coordination & Administration Remote
 - Directed team registration, administrative workflows, and official communications with global organizers.
 - Managed cross-functional coordination with Task Leaders across multiple competition tracks to ensure seamless participation and technical execution.
- **Research Intern — MIT Lab, SMST, IIT Kharagpur** Jan 2025 – Jul 2025
Deep Learning for OPMD Classification Kharagpur, India
 - Engineered deep learning architectures to analyze Oral Cytology Whole Slide Images (WSIs) for the automated classification of Oral Potentially Malignant Disorders (OPMD).
 - Collaborated closely under the supervision of Dr. Subhamoy Mandal and Dr. Shirin Dasgupta to bridge computational models with clinical pathology needs.
- **Teaching Assistant — SMST, IIT Kharagpur** Jul 2025 – Apr 2026
Courses: Introduction to Programming, Pathological Image Processing Kharagpur, India
 - Instructed MMST students (medical professionals/MBBS doctors) in Python programming, focusing on algorithmic thinking and computational problem-solving.
 - Mentored graduate students in developing Python-driven image analysis workflows specifically tailored for pathological applications.

Thesis & Selected Projects

- **M.Tech Thesis — IIT Kharagpur** *Jan 2025 – Apr 2026*
Physics-Aware Learning for Modulo Sinogram Unwrapping & Reconstruction
 - Engineered a physics-informed machine learning framework to accurately unwrap modulo sinograms, directly integrating CT physics and forward-projection models into the learning pipeline.
 - Formulated and solved complex inverse problems by embedding physical imaging constraints into deep learning architectures, systematically reducing high-density artifacts and noise.
 - Enhanced the overall robustness and spatial resolution of CT image reconstructions, outperforming traditional analytical methods.
- **B.Tech Thesis — KGEC** *2023 – 2024*
Pancreatic Cancer Prediction from Imaging & Biomarkers
 - Developed Convolutional Neural Networks (CNNs) to automate the detection of pancreatic tumors from CT scans.
 - Implemented machine learning classifiers on clinical urinary biomarkers, surpassing previous state-of-the-art predictive accuracy by 0.7%.

Technical Skills

- **Programming:** Python, C/C++, MATLAB
- **ML/DL:** PyTorch, MONAI, scikit-learn, Optuna
- **Medical Imaging:** OpenCV, Pillow, SimpleITK, NiBabel, PyRadiomics, ITK
- **Data/Visualization:** NumPy, Pandas, Matplotlib, Seaborn, SciPy
- **MLOps/Tools:** Git, Docker, Linux, HPC

Achievements

- **AIR 2741** (Top 2.5%) among 120K+ candidates in GATE Computer Science (2024).
- **GMR 1801** (Top 1.8%) among 100K+ candidates in WBJEE (2020).
- **Olympiad Rank 7** in iOS & iOM (SilverZone, 2013).
- **10/10 CGPA** in CBSE Secondary School Examination (2017).

Summer Schools & Academic Training

- **JTG / IEEE IT Society Summer School** *Jul 2025*
Department of Electrical Engineering, C-MInDS, IIT Bombay
 - Completed advanced training in Deep Learning, Graph Signal Processing, and Wireless Networks.
 - Engaged with pioneering lectures and invited talks from leading researchers, including faculty from UCSD and Georgia Tech.
- **IITB SysCon Summer School** *May 2023*
Department of Systems & Control, IIT Bombay
 - Participated in intensive theoretical sessions covering Quantum Control, Multi-Agent Systems, Reinforcement Learning, and Vibrational Control.